Quorum sensing & *Agr* interference in *S. aureus*

Study guide — companion to the 48-second, five-chapter teaching animation of how staphylococci count themselves with AIP-1, and how a foreign signal can silence their quorum.

5 chapters · 9.6 s each · seamless loop (frame 0 ≡ frame 48)

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How to use this guide. Watch the animation once in full, then read each section below while scrubbing back to the matching chapter (the five dots under the play button jump straight to a chapter; `?t=seconds` in the URL freezes any frame). The questions at the end test exactly what the film shows — try each one before clicking it open for its answer. Where the animation takes teaching liberties, they are flagged openly so you do not mistake them for scale.

1 The five-chapter loop at a glance

Chapter	Starts	What you see on screen	Idea to take away
1 · SILENCE	0 s	A sparse culture. Each coccus makes AIP-1, but at low density the peptide is diluted below the threshold (dashed line on the gauge); *AgrC* stays unphosphorylated — red ✕ over P2/P3; X-gal dish colourless.	Signal + receiver exist at every density. What changes with density is the **concentration**, and only above threshold does anything happen.
2 · GROWTH & ACCUMULATION	9.6 s	Cocci divide on the left; each daughter adds its own AIP-1 share and the blue meter climbs. Mid-chapter the threshold crossing flashes orange: *AgrC* hands a phosphate to *AgrA*, which fires P3.	Growth is what raises concentration — every division event injects more signal into the shared medium. The switch flips at a defined threshold, not gradually.
3 · QUORUM ON	19.2 s	The autocatalytic loop: amber *agrBDCA* mRNA marching along the operon; AgrA~P docked under P2/P3. Inside the cell, the *AgrD* propeptide is cleaved and cyclised by *AgrB* into mature AIP-1 that streams out into the population. The dish turns blue.	Above threshold the system **amplifies itself**: more signal → more receptor/signal genes → even more signal. Virulence output (blue = P3-*lacZ*) is a population-level all-or-none event.
4 · INVASION → SILENT	28.8 s	A skin commensal slides in on the right, multiplies first (three daughters), then floods the medium with non-cognate red AIP. It displaces AIP-1 at *AgrC*: P3 stays off despite high density — colonies stay colourless.	Interference is **competitive antagonism, not killing**. A foreign peptide that binds the receptor but does not activate it can veto quorum even in a dense culture.

5 · RECOVERY & RESET	38.4 s	The foreign signal washes out; cognate AIP-1 re-binds and the loop refires (blue returns). Then dilution — growth into fresh medium — pushes AIP below threshold again: back to silence, exactly where we started.	Both states are **reversible**. Quorum is not a permanent identity; it tracks current signal concentration in real time.

2 Quorum sensing: the counting mechanism

Quorum sensing (QS) is a population-level communication system: cells secrete small signalling molecules called **autoinducers**, and when their extracellular concentration crosses a threshold, the whole population co-ordinately switches on genes that are only useful in numbers. In *Staphylococcus aureus* this is the *agr* (accessory gene regulator) system.

The signal: AIP-1, a thiolactone peptide

AIP-1 (autoinducer peptide 1) is an octapeptide — sequence **YSTCDFIM**. Its N-terminal Tyr-Ser-Thr tail projects from a five-residue ring (**Cys^4–Asp^5–Phe^6–Ile^7–Met^8**) closed by a thiolactone S-bond. The ring is not decoration: it is essential for receptor activity, and even single amino-acid differences in the ring separate the AIP groups (Section 4).

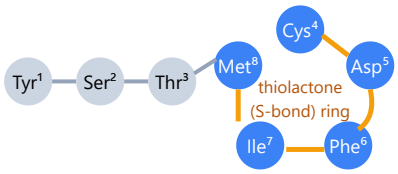


Figure A. AIP-1 = YSTCDFIM: a Tyr-Ser-Thr tail (grey) attached to the Cys^4–Met^8 thiolactone ring (blue, amber bonds). The ring is essential for receptor activity; one residue change (Cys→Tyr at position 6) gives AIP-4.

Making and exporting the signal

The *agrD* gene encodes a ~45-residue **propeptide**, which is biologically inert. The transmembrane protein **AgrB** — a cysteine endopeptidase anchored in the membrane — cleaves the C-terminus of the propeptide and cyclises it into the thiolactone; this same step gates export, so mature AIP-1 appears directly outside the cell. That is why the film shows the ring travelling out through AgrB: **maturation and secretion are one coupled event**.

The receiver: a two-component system

AgrC spans the membrane (a histidine-kinase receptor). When AIP-1 binds its extracellular domain, AgrC transfers a phosphate to **AgrA**, the response regulator in the cytoplasm. Phosphorylated **AgrA~P** is the transcription factor that actually acts on DNA: it turns on P3 (virulence genes) and — crucially — re-switches on its own operon at P2, closing a positive-feedback loop.

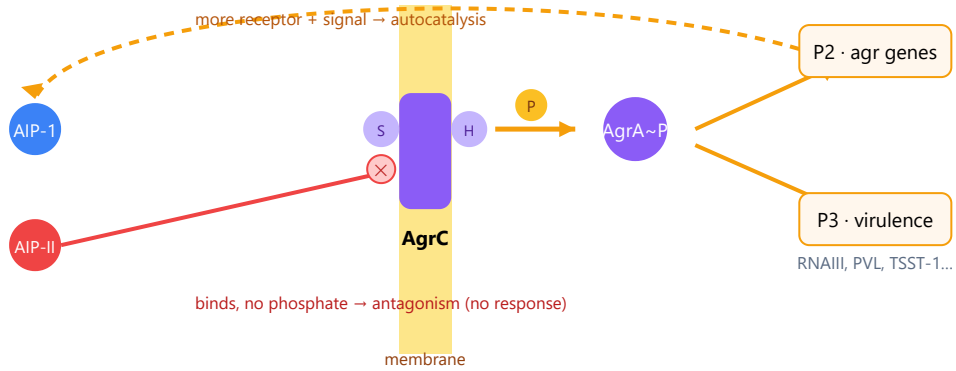


Figure B. The AgrC–AgrA two-component system. Blue AIP-1 triggers phosphotransfer to AgrA, which activates P3 (virulence) and re-activates its own operon at P2 (autocatalytic loop). Red non-cognate AIP occupies the receptor without triggering phosphorylation — antagonism, not an opposite response.

The threshold logic

- **Below threshold:** AgrC is mostly unphosphorylated; P2/P3 are off (the red ✕ marks in the film). The culture can still make AIP-1 — production and sensing run at every density.
- **Crossing:** a sharp, population-wide flip — not a slow ramp. In the film this is the orange threshold flash.
- **Above threshold (autocatalysis):** AgrA~P re-switches on its own operon at P2, so more receptor and signal are made; quorum reinforces itself until concentration falls again.

> **How this compares with Gram-negative systems.** In the better-studied Gram-negative examples (for instance *Vibrio fischeri*, the classic model), the autoinducer is a small lactone — an acyl-homoserine lactone, or AHL — that simply diffuses back into cells and switches on its own genes. Staphylococci take a different route: their signal is a processed **peptide**, and detection requires the dedicated membrane receptor AgrC with phosphotransfer to AgrA — the two-component system in Figure B. Same population-counting logic, different chemistry and machinery.

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3 The output: what "QUORUM ON" actually turns on

P3 drives a regulon centred on the regulatory RNA **RNAIII**, which in turn activates virulence factors and represses protective ones. Two exotoxins named in the animation are real, Agr-dependent proteins:

- **PVL (leucocidin)** / lukSF-PV — kills neutrophils.
- **TSST-1** (toxic shock syndrome toxin-1) / C3 — superantigens behind toxic-shock physiology.

In many other species, the quorum output is not virulence but **biofilm formation** — dense, surface-attached communities that resist antibiotics and disinfectants. The film focuses on the virulence branch because it is what makes the *S. aureus* story clinically relevant; the counting logic itself (Section 2) is shared with those biofilm-building systems.

A note before reading the dish: *Staphylococcus* does not normally make β-galactosidase — that enzyme belongs to *E. coli* (the *lacZ* gene) — and no wild staphylococcal strain carries RNAIII fused to *lacZ*. The blue dish in the film shows an **engineered reporter strain**: researchers inserted a copy of *lacZ* under control of the Agr-dependent P3 promoter (for example, strain SH101F7, where the reporter sits at the RNAIII locus as an RNAIII::*lacZ* fusion). The cell therefore makes β-galactosidase only when quorum has switched P3 on. On plates containing X-gal — a colourless galactose derivative that β-galactosidase cleaves into an insoluble blue product — colonies with active quorum turn blue; colonies with the switch off stay colourless. So in the

film, **blue = quorum ON; colourless = off**, whether from low density or from interference. That ambiguity is itself a teaching point: the readout reports switch state, not cause.

> **Why this reporter was built (Canovas et al. 2016, Fig 1).** The P3-*lacZ* construct is a screening tool for finding **skin commensals that block *S. aureus* quorum sensing**. The reporter strain is streaked on a plate and a candidate skin bacterium is spread beside it as a lawn. Colonies far from the lawn turn blue — their quorum works normally. But if the commensal secretes an interfering AIP or some other molecule that blocks staphylococcal quorum sensing, the reporter colonies growing near it stay **colourless even at high density**: P3 was never switched on, so no β-galactosidase is made. Colourless does not mean dead or sick — it means P3 is off. Commensals whose secretions silence staphylococcal quorum are identified precisely by this colourless halo.

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4 Agr interference: non-cognate AIPs and the AIP groups

Clinical *S. aureus* isolates fall into four **AIP groups (I–IV)**, defined by their autoinducer peptide sequence; each group's AgrC responds best to its own ("cognate") AIP. Approximate distribution among clinical isolates: I ≈ 60 %, II ≈ 22 %, III ≈ 14 %, IV ≈ 2.5 %.

Group	Share of isolates	Cognate AIP	Note
I	~60 %	AIP-1 = YSTCDFIM	The "standard" strain; the film's protagonist.
II	~22 %	AIP-II (different ring residues)	The film's commensal signal, drawn in red. Classic source of interference against group I strains.
III	~14 %	AIP-III	Frequently found in livestock-associated lineages.
IV	~2.5 %	AIP-4 = YSTCYFIM (Cys^{6}→Tyr)	Rare; the "I & IV interchangeable" note on screen is slide-level shorthand — groups I and IV share the closest sequences but are not functionally identical in every assay.

Mechanism of interference: a non-cognate AIP that binds an alien AgrC occupies it without triggering phosphotransfer. Cognate AIP-1 is competitively displaced, so even at high density the two-component switch never fires — P3 stays off. This is **antagonism**: the foreign signal produces "no response", not a reversed one, and it does not kill the host cells.

> **The same trick happens on real skin (Paharik et al. 2017).** The coagulase-negative staphylococci that live on us as commensals keep a colonising *S. aureus* in check: their non-cognate signals stop the pathogen from switching its virulence regulon on, even while it is established there. That is chapter 4's antagonism observed in nature rather than on an agar plate.

Cross-group inhibition is dose-dependent and graded: close relatives still inhibit each other (measured EC_{50}s, Gordon et al. 2013), so "non-cognate" is a spectrum, not a wall. And because binding is reversible, when the foreign signal clears — as in chapter 5 — cognate AIP-1 re-binds and the loop refires.

5 Where the animation is simplified

- **Scale:** cocci (~1 μm) and AIP monomers (~1 kDa) are drawn huge; at true scale neither would be visible.
- **Counts vs concentration:** the ten blue / up-to-five red glyph molecules stand for *concentration*, not molecule counts or cell numbers; the threshold meter is a teaching device, not an assay readout.
- **One cell tells the population's story:** real QS is population-level — the marching mRNA and export ring in chapter 3 stand in for what every above-threshold cell does at once.
- **The commensal** is a generic coagulase-negative staphylococcus making non-cognate (e.g. group II) AIP; red shows competitive antagonism, not killing — the *S. aureus* population survives and refires when the foreign signal clears.
- **Time is compressed evenly:** each chapter gets 9.6 s regardless of how long that state lasts in reality (a real culture can hold quorum for hours, or never reach it).

6 Vocabulary quick-reference

Term	Meaning in this system
Quorum sensing	Density-dependent co-ordinated gene regulation via secreted autoinducers.
Autoinducer (AIP)	The small signalling peptide; AIP-1 = YSTCDFIM thiolactone in group I strains.
Cognate / non-cognate	AIP matching the strain's own AgrC vs a foreign one that binds without activating it.
Thiolactone ring	Cys^{4}–Met^{8} macrocycle closed by an S-bond; essential for receptor activity.
Propeptide (AgrD product)	The ~45-residue inert precursor that AgrB cleaves and cyclises into mature AIP, coupled to export.
AgrB	Transmembrane cysteine endopeptidase; maturation + exporter for AIP-1.
Two-component system	Membrane sensor kinase (AgrC) + cytoplasmic response regulator (AgrA), coupled by phosphotransfer.
AgrA~P	Phosphorylated AgrA — the transcription factor that activates P3 and re-activates its own operon at P2.
P2 / P3	P2 = autocatalytic *agr* transcription; P3 = virulence regulon (RNAIII, exotoxins).
Autocatalysis / positive feedback	The output of the switch (more receptor + signal) amplifies its own input — quorum reinforces itself.
Threshold	The AIP concentration at which AgrC–AgrA flips on; density-dependent because signal is shared in the medium and diluted when sparse.
RNAIII	Regulatory RNA under P3 that activates virulence genes (e.g. PVL, TSST-1) and represses others.
P3-lacZ / X-gal readout	An **introduced reporter, not native to staphylococci**: an engineered copy of *lacZ* under P3 control makes β-galactosidase only when quorum is ON; on X-gal plates blue colonies = P3 active.
Competitive antagonism	Non-cognate AIP occupying the receptor and vetoing quorum without activating or killing — "no response", not an opposite one.

AIP groups I–IV	The four autoinducer families of clinical *S. aureus* (~60/22/14/2.5 %); basis of cross-strain interference.

7 Check your understanding

Q1. Why does AIP-1 accumulate only when the population is dense, even though every cell — sparse or crowded — is making it?

A1. AIP-1 is secreted into the shared medium, so its concentration scales with cell number per unit volume. When sparse it diffuses away and stays below threshold; as divisions add producers (and each daughter inherits production), the cumulative concentration crosses threshold and flips AgrC–AgrA on.

Q2. What structural feature makes AIP-1 biologically active, and which protein creates it? Why are maturation and export described as one coupled event?

A2. The Cys^{4}–Met^{8} thiolactone ring — essential for receptor activity. *AgrB*, a transmembrane cysteine endopeptidase, cleaves the ~45-residue AgrD propeptide and cyclises it; that same step gates export through the membrane, so mature AIP-1 appears directly extracellularly.

Q3. Name the four proteins of the *agr* locus shown on screen and state what P2 versus P3 each control.

A3. *agrA* (response regulator), *agrC* (receptor/histidine kinase), *agrD* (signal propeptide) and *agrb* (maturation/export endopeptidase). P2 drives transcription of the signal genes themselves (autocatalysis); P3 drives virulence genes including RNAIII.

Q4. Which molecule actually acts as the transcription factor, and in which chemical form?

A4. AgrA, only in its phosphorylated form (AgrA~P), after phosphotransfer from AIP-1-bound AgrC.

Q5. Why is the *agr* circuit called an "autocatalytic loop"? What keeps it from running away indefinitely?

A5. Because the output — more receptor and more signal production at P2 — amplifies the input, a positive-feedback loop. It stops when concentration falls: dilution by growth into fresh medium (or loss of cells) drops AIP below threshold and the switch reverses; nothing is permanently "on".

Q6. On a P3-lacZ / X-gal plate, what does blue report — and why can't the colour alone tell you *why* the switch is on or off?

A6. Blue means P3 is active, read out through an **introduced reporter**: staphylococci do not natively make β-galactosidase — in these strains it comes from a copy of *lacZ* inserted under P3 control. The colour cannot distinguish quorum-on-from-density (chapter 3) from re-firing after recovery — and it reads identically colourless for "too few cells" vs "interfered", so the cause needs other evidence.

Q7. A skin commensal floods the medium with group II AIP while a dense group I culture sits beside it. Predict the X-gal colour of the *S. aureus* colonies, and explain at the level of AgrC–AgrA why that happens despite high density.

A7. Colourless. Group II AIP binds group I AgrC as a competitive antagonist: occupancy without phosphotransfer, plus displacement of cognate AIP-1 — so AgrA is never phosphorylated and P3 stays off even though the culture is dense. Interference is antagonism, not lysis.

Q8. List the four AIP groups with their approximate shares among clinical isolates, and give the sequences of AIP-1 and AIP-4. What is the single-residue difference?

A8. I ≈ 60 %, II ≈ 22 %, III ≈ 14 %, IV ≈ 2.5 %. AIP-1 = YSTCDFIM; AIP-4 = YSTCYFIM — one ring residue, Cys^{6}→Tyr (position 6).

Q9. After the foreign signal washes out, why do the colonies turn blue again without any new gene being made?

A9. The system is fully reversible: the *agr* machinery was never destroyed, only out-competed at the receptor. When foreign AIP clears, cognate AIP-1 re-binds AgrC, phosphotransfer resumes, and the still-intact autocatalytic loop refires.

Q10. In chapter 5, dilution by growth into fresh medium resets quorum. Explain why dilution — not cell death — is what matters here.

A10. Quorum tracks concentration in the shared medium. Growth into fresh volume lowers AIP per unit volume below threshold even though every cell is alive and producing — so AgrC dephosphorylates, P3 switches off, and the population returns to the silent state of chapter 1 (the film's seam).

8 References

Simplified for teaching; species-specific facts verified against primary literature. Links are stable PubMed/PMC identifiers.

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Companion animation: `index.html` in this repository (or the GitHub Pages site) · deterministic 48 s loop, scrubbing and freeze-frames via URL parameters.